Newton’s Method (Lec9.pdf)

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Todays topics & Logistics

- Newton method
  - The direction
  - The step size
  - The iterates (may not converge, increase the objective, but are fast…)

Gradients shop locally, Newton shops globally

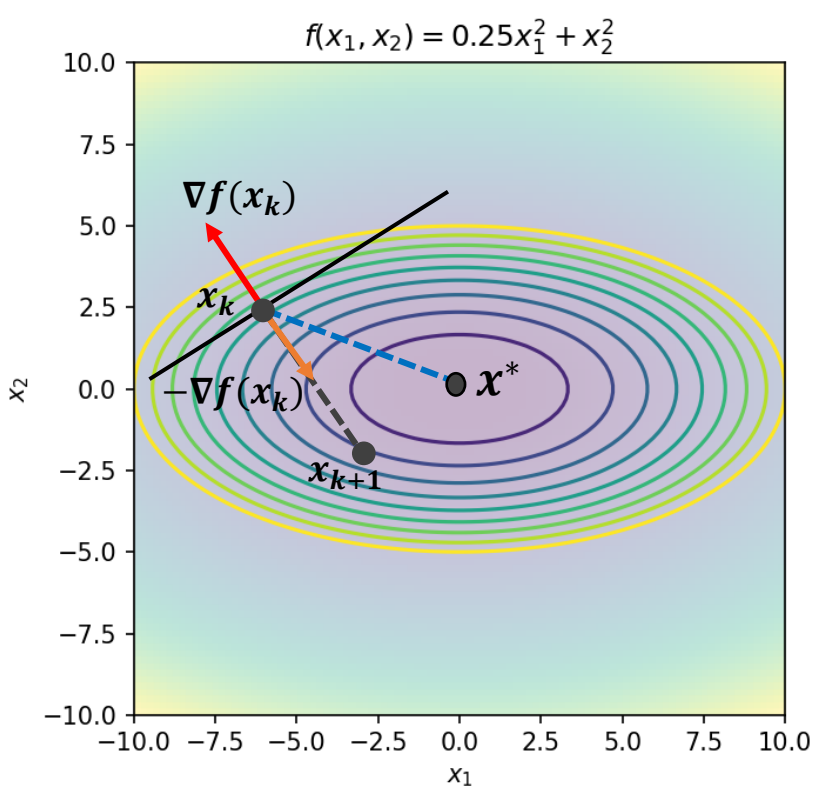
- Gradient descent methods use **1st order information** (i.e., gradients) to determine the next iteration.

- **Locally,** $-\nabla f(x_k)$ gives us the direction of maximum rate of decrease, but **globally** it may not be the best direction.

- For our example we have $-\nabla f(x) = -(0.5x_1, 2x_2)^T$

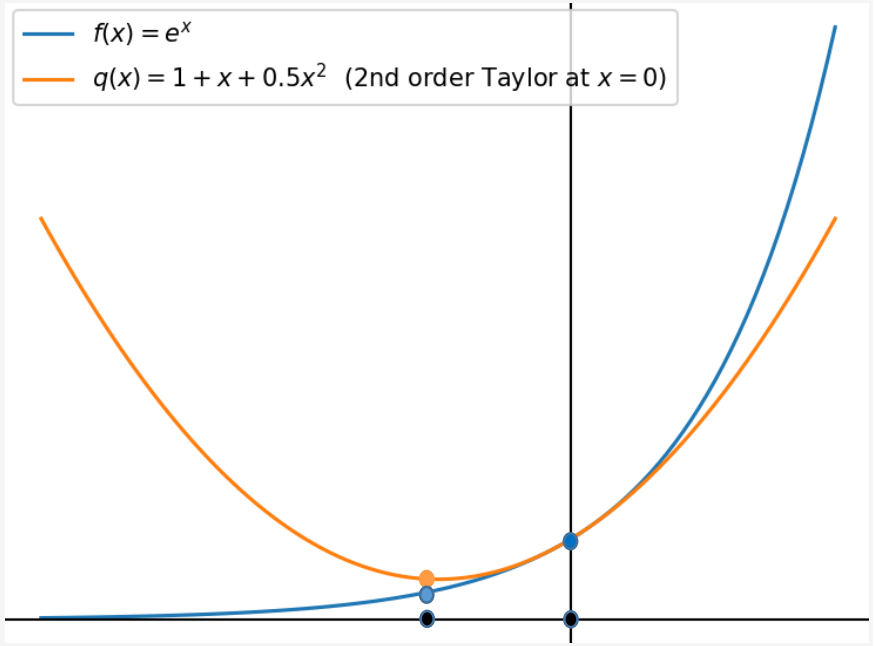
  - **Not well scaled** and therefore it may not easily see the global information (i.e., where the optimal solution x^* may reside…)

- **Newton's method** is doing better job in identifying the **global information** and may reach the optimum faster!



Motivation for Newton's method

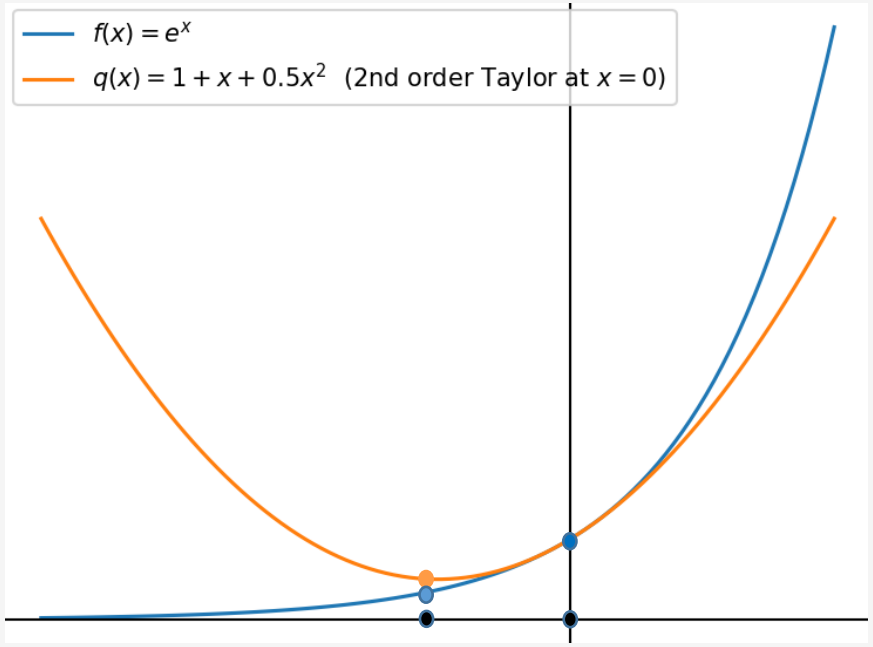
- Recall what we did in $n = 1$ (one dimension).
  - Approximated the function f by a quadratic
- We do the same when $n \geq 2$
- **Question:** What is the best and easiest **quadratic** approximation of a **multivariate** nonlinear function at a random point $x_k \in \boldsymbol{R^n}$??



$x_{k+1} = -1 \quad x_k = 0$

Motivation for Newton's method

- Recall what we did in $n = 1$ (one dimension).
  - Approximated the function f by a quadratic
- We do the same when $n \geq 2$
- The 2nd order Taylor approximation of f at $x_k \in R^n$.



$x_{k+1} = -1 \quad x_k = 0$

$$q(x) \approx f(x_k) + \nabla f(x_k)^T (x - x_k) + \frac{1}{2}(x - x_k)^T \nabla^2 f(x_k)(x - x_k)$$

Motivation for Newton's method

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- **Important questions:**
  - Is q bounded from below?
  - How should we determine the minimum of q(x)?

Motivation for Newton's method

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- **Important questions:**
  - Is q bounded from below?
  - How should we determine the minimum of q?
- If $\nabla^2 f(x_k) \succ 0$, then q has a **unique** minimum $\hat{x}$
- Newton method chooses that **unique minimum** $\hat{x}$ as the **next iterate** x_{k+1}

q is a quadratic approximation of f at $f(x_k)$

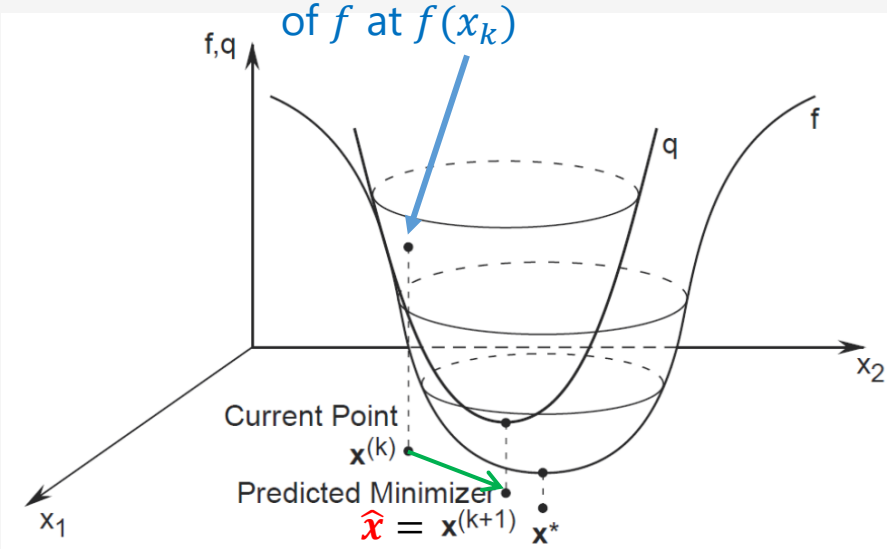

Image taken from: Chong & Zak [CZ13]: "An Introduction to Optimization", 2013

Compute x_{k+1}

$$q(x) \approx f(x_k) + \nabla f(x_k)^T (x - x_k) + \frac{1}{2}(x - x_k)^T \nabla^2 f(x_k)(x - x_k)$$

- Use **FONC**: $\nabla q(\hat{x}) = 0 \Rightarrow \nabla f(x_k) + \nabla^2 f(x_k)(\hat{x} - x_k) = 0 \Rightarrow$

$$\Rightarrow \nabla^2 f(x_k)\hat{x} = \nabla^2 f(x_k) x_k - \nabla f(x_k) \overset{\nabla^2 f(x_k) \succ 0}{\Rightarrow} \hat{x} = x_k - \left(\nabla^2 f(x_k)\right)^{-1} \nabla f(x_k)$$

- Now set $x_{k+1} = \hat{x} \quad \Rightarrow \quad x_{k+1} = x_k - \left(\nabla^2 f(x_k)\right)^{-1} \nabla f(x_k)$

- x_{k+1} exists and is unique because $\nabla^2 f(x_k) \succ 0$

- **Question: What happens if $\nabla^2 f(x_k) \succcurlyeq 0$??**

Recall that
$d_k = -\left(\nabla^2 f(x_k)\right)^{-1} \nabla f(x_k)$
is a **descent direction**, when
$\nabla^2 f(x_k) \succ 0$

Compute x_{k+1}

$$q(x) \approx f(x_k) + \nabla f(x_k)^T (x - x_k) + \frac{1}{2}(x - x_k)^T \nabla^2 f(x_k)(x - x_k)$$

- **What happens if $\nabla^2 f(x_k) \succcurlyeq 0$??**
- The Hessian $\nabla^2 f(x_k)$ has **at least one zero eigenvalue** ⇒ Not invertible!
- **Newton may break (stop) in such a case**

Compute x_{k+1}

$$q(x) \approx f(x_k) + \nabla f(x_k)^T (x - x_k) + \frac{1}{2}(x - x_k)^T \nabla^2 f(x_k)(x - x_k)$$

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- The Hessian $\nabla^2 f(x_k)$ has **at least one zero eigenvalue** ⇒ Not invertible!
- **Newton may break (stop) in such a case**
- **To keep Newton going** we can add $\lambda > 0$ to its diagonal
  - In a way we consider the modified quadratic approximation:

$$q_\lambda(x) \approx f(x_k) + \nabla f(x_k)^T (x - x_k) + \frac{1}{2}(x - x_k)^T (\nabla^2 f(x_k) + \lambda I)(x - x_k) \quad \Rightarrow \quad \underline{x_{k+1} = x_k - (\nabla^2 f(x_k) + \lambda I)^{-1} \nabla f(x_k)}$$

Compute x_{k+1}

$$q(x) \approx f(x_k) + \nabla f(x_k)^T (x - x_k) + \frac{1}{2}(x - x_k)^T \nabla^2 f(x_k)(x - x_k)$$

- What happens if $\nabla^2 f(x_k) \succcurlyeq 0$??
- The Hessian $\nabla^2 f(x_k)$ has **at least one zero eigenvalue** $\Rightarrow$ Not invertible!
- **Newton may break (stop) in such a case**
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What happens if $\nabla^2 f(x_k)$ is indefinite ??

- In a way we consider the modified quadratic approximation:

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Compute x_{k+1}

$$q(x) \approx f(x_k) + \nabla f(x_k)^T (x - x_k) + \frac{1}{2}(x - x_k)^T \nabla^2 f(x_k)(x - x_k)$$

- What happens if $\nabla^2 f(x_k)$ is indefinite ?? $\rightarrow$ $\boldsymbol{d_k = -\left(\nabla^2 f(x_k)\right)^{-1} \nabla f(x_k)}$ is **NOT** descent direction
- The Hessian $\nabla^2 f(x_k)$ has **at least one positive and one negative eigenvalue**
- **Newton may not point to a point with smaller objective function value (will not decrease)**
- **To keep Newton going** we can add $\lambda > |\lambda_{min}|$ to its diagonal (λ_{min} = smallest eigenvalue)
  - In a way we consider the modified quadratic approximation:

$$q_\lambda(x) \approx f(x_k) + \nabla f(x_k)^T (x - x_k) + \frac{1}{2}(x - x_k)^T (\nabla^2 f(x_k) + \lambda I)(x - x_k) \quad \Rightarrow \quad x_{k+1} = x_k - (\nabla^2 f(x_k) + \lambda I)^{-1} \nabla f(x_k)$$

Levenberg-Marquardt method

The Newton algorithm (pseudocode)

- **Step 0:** Select an initial point x_0, set $k = 0$, and choose a small $\epsilon > 0$
- **Step 1:** Calculate a descent direction $d_k = -\left(\boldsymbol{\nabla^2 f(x_k)}\right)^{-1} \nabla f(x_k)$;
- **STEP 2:** If $\left|\left|\nabla f(x_k)\right|\right| \leq \epsilon$ Then STOP → x_k is a candidate optimal solution
- **Step 3:** Choose step size α_k
- **Step 4:** Find next iterate: $x_{k+1} = x_k + \alpha_k d_k$
- **Step 5:** Set $k = k + 1$ and GOTO **Step 1**

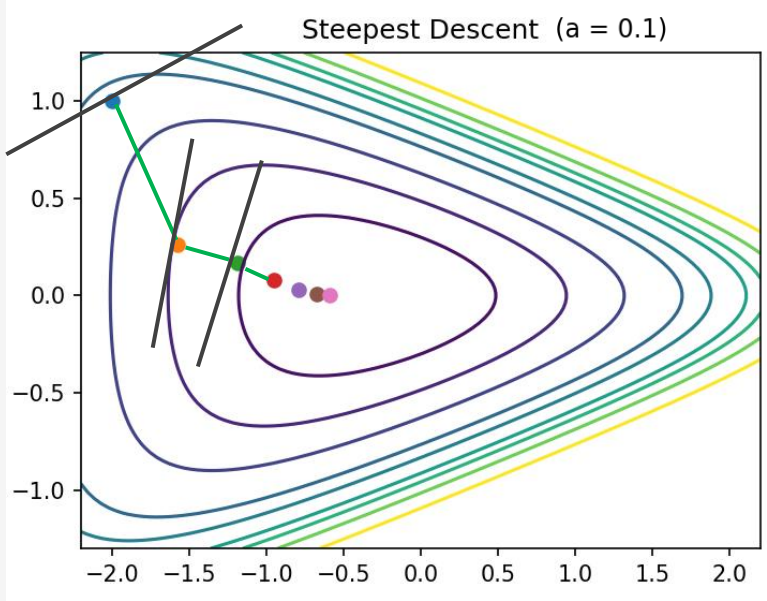
Computationally expensive task

Add λI if needed

Comparison of Steepest Descent and Newton Methods

Example: $\min_{x \in R^2} f(x_1, x_2) = e^{x_1+3x_2-0.1} + e^{x_1-3x_2-0.1} + e^{-x_1-0.1}$, with $x_0 = (-2,1)^T$

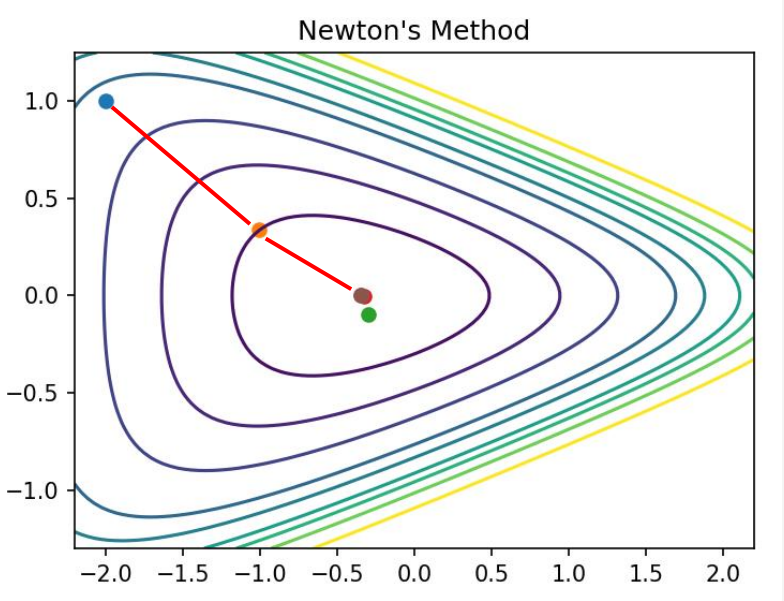
$x^* = (0.346,0)^T$



In 7 iterations:

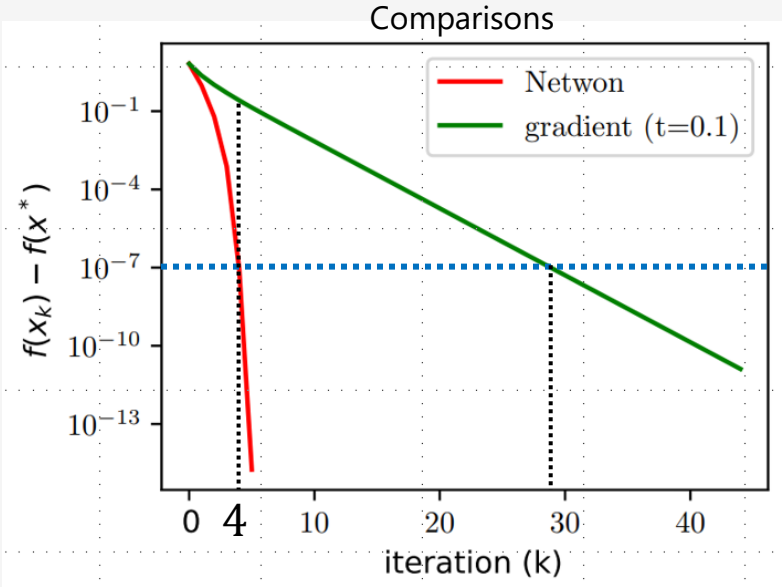
$|f(x_7) - f(x^*)| = 0.625$

$x^* = (0.346,0)^T$



In 5 iterations:

$|f(x_5) - f(x^*)| = 1.1 \times 10^{-7}$



However,
Newton iterations are more expensive

Convergence of Newton method (in a vicinity of $\boldsymbol{x}^*$)

- **Theorem:** Suppose
  - $f: R^n \to R$ is $\boldsymbol{C^3}$ (i.e., three times continuously differentiable),
  - for a point $x^* \in R^n$ we have $\nabla f(x^*) = 0$, and
  - $\nabla^2 f(x^*)$ is invertible.
- Then there exists an $\epsilon > 0$ such that the Newton iterations **starting from any point** $\mathbf{x_0} \in \boldsymbol{B(x^*, \epsilon)}$ are **well defined** and **converge** to the point $x^* \in R^n$.
- The **convergence rate is quadratic** → $||x_{k+1} - x^*|| \leq ||x_k - x^*||^2$

Convergence of Newton method (in a vicinity of $\boldsymbol{x}^*$)

- **<u>Interpretation</u>**: Newton's method will converge very fast from any starting point x^0 as long as $\mathbf{x_0} \in \boldsymbol{B}(\boldsymbol{x}^*, \boldsymbol{\epsilon})$, **that is, it is convergent as long as it starts close enough to a critical point (i.e., satisfying FONC).**

- The convergent point x^* may be a local minimum or a local maximum, or saddle point.

- To ensure that x^* is a local minimum, check whether $\nabla^2 f(x^*) \succ 0$

- If $\boldsymbol{x}^0$ **is far away from** $\boldsymbol{x}^*$ then there is **no guarantee** that the Newton method will converge if the function f is **nonconvex**.

  - **It may not be descent direction** → when $\boldsymbol{\nabla}^2\mathbf{f}(\mathbf{x_k}) \not\succeq \mathbf{0}$ we may have: $f(x_{k+1}) > f(x_k)$

Newton method with step size – Armijo rule

The Newton direction is always descent for convex functions.

<u>Observation</u>: Any function within at a small neighborhood can be approximated by a convex function

Hence if you make a **small enough step** $\boldsymbol{\alpha_k} \in (0,1]$ along the Newton direction you will guarantee descent → $f(x_{k+1}) < f(x_k)$

$$x_{k+1} = x_k - \alpha_k \left(\nabla^2 f(x_k)\right)^{-1} \nabla f(x_k)$$

The Newton algorithm with Armijo rule (pseudocode)

- **Step 0:** Select an initial point x_0, set $k = 0$, and choose a small $\epsilon > 0$
- **Step 1:** Calculate a descent direction $d_k = -\left(\boldsymbol{\nabla^2 f(x_k)}\right)^{-1} \nabla f(x_k)$;
- **STEP 2:** If $\left|\left|\nabla f(x_k)\right|\right| \leq \epsilon$ Then STOP → x_k is a candidate optimal solution
- **Step 3:** Choose step size α_k
- **Step 4:** Find next iterate: $x_{k+1} = x_k + \alpha_k d_k$
- **Step 5:** Set $k = k + 1$ and GOTO **Step 1**

Computationally expensive task

Add λI if needed

Armijo rule:

3.1: Set $j = 0$ and $a_j = 1$

3.2: If $\boldsymbol{f\left(x_k + \alpha_j d_k\right) \leq f(x_k) - \epsilon\, \alpha_j\, \nabla f(x_k)^T d_k}$

then STOP → return $\alpha_k = \boldsymbol{\alpha_j}$

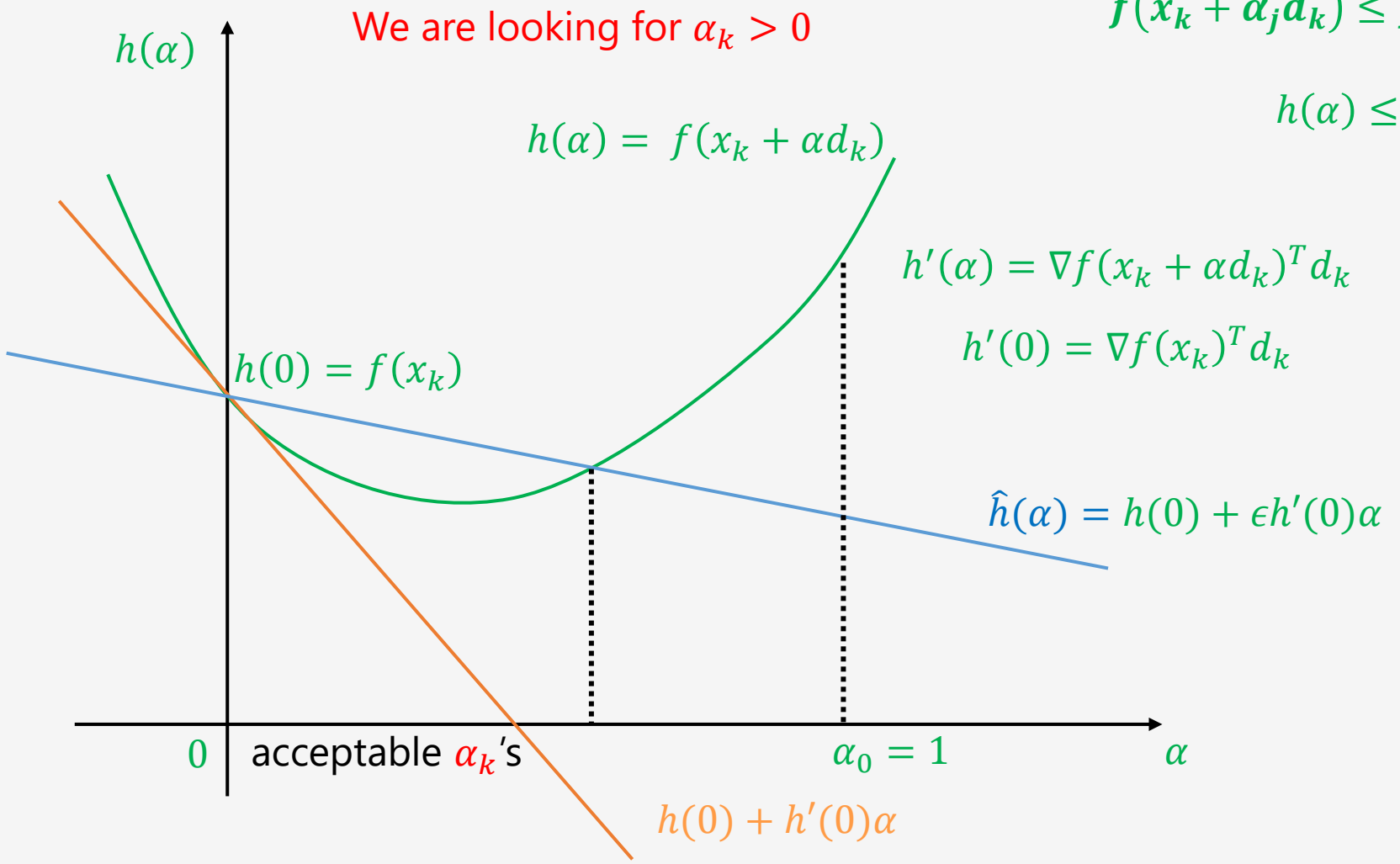
Else reduce the step: $a_{j+1} = \frac{1}{\delta}\alpha_j$ where $\delta > 1$

3.3: Set $j = j + 1$ and GOTO step 3.2

Armijo rule – geometric representation

$$\boldsymbol{f(x_k + \alpha_j d_k) \le f(x_k) - \epsilon\, \alpha_j\, \nabla f(x_k)^T d_k}$$

$$h(\alpha) \le h(0) + \epsilon h'(0)$$

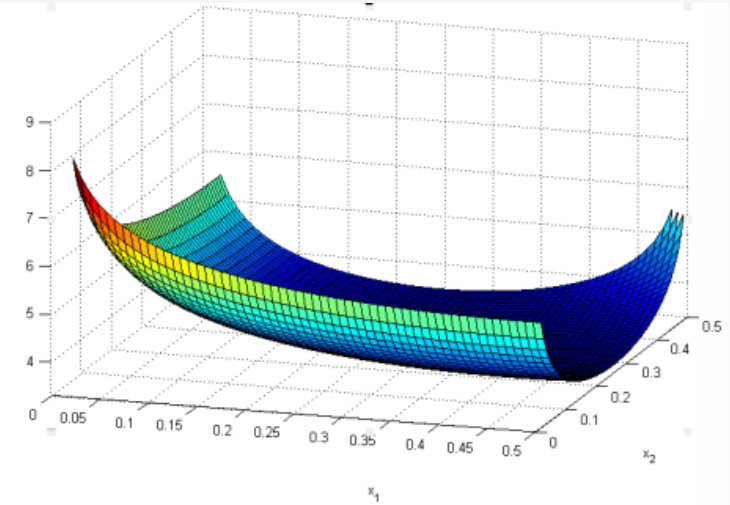


Pseudocode of Newton method (Matlab – from the notes)

```
syms x1 x2;
f=-log(1-x1-x2)-log(x1)-log(x2);
gradf=gradient(f);
Hf=hessian(f);
N=10; %number of Newton iterations
x=zeros(2,N); fval=zeros(N,1); er=zeros(N,1);
x(:,1)=[.8;.1]; %initial point

for k=1:N-1
   fval(k)=subs(f,{x1,x2},{x(1,k),x(2,k)});
   er(k)=norm([1/3;1/3]-x(:,k));
   x(:,k+1)=x(:,k)- subs(Hf,{x1,x2},{x(1,k),x(2,k)})\
subs(gradf,{x1,x2},{x(1,k),x(2,k)});
end
 fval(k+1)=subs(f,{x1,x2},{x(1,k+1),x(2,k+1)});
 er(k+1)=norm([1/3;1/3]-x(:,k+1));
format long
output=[(1:N)' x' er fval]
```

$$f(x) = -\log(1 - x_1 - x_2) - \log x_1 - \log x_2$$



If we select $x^0 = (0.8, 0.1)$, then Newton method will converge with $\alpha = 1$ to the optimal solution:

$x^* = \left(\frac{1}{3}, \frac{1}{3}\right)$ and $f(x^*) = 3.295836$.

For points far away from x^* use Armijo rule.